(1) Use separate Venn Diagrams and shade the regions for the following:

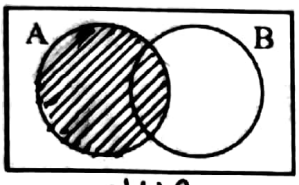
(a) A (b) B' (c) A ∪ B (d) A'∪ B

(e) A ∪ B' (f) A'∪ B' (g) (A ∪ B)' (h) (A'∪ B)'

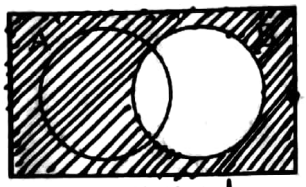
(i) A ∩ B (j) A'∩ B (k) A ∩ B ' (l) A'∩B'

(m) (A ∩ B)' (n) (A ∪ B) ∪ (A'∪ B ')

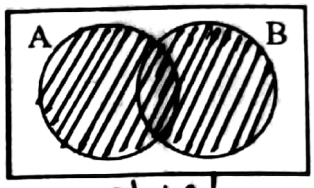
**Solution :**




~~A'UB~~ A

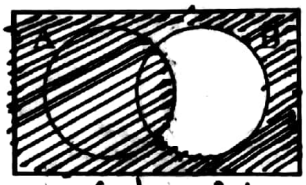


~~AUB'~~ B'

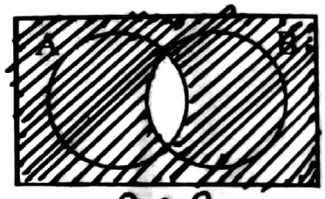



~~A'UB'~~ AUB

~~(AUB)'~~ A'UB

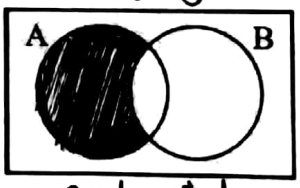


~~(A'UB)'~~ AUB'

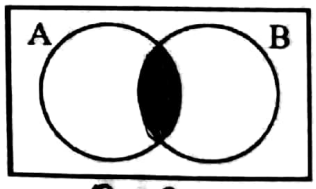


~~AnB~~ A'UB'

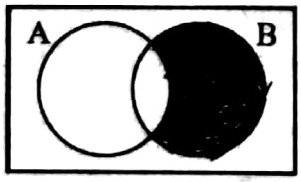
(AUB')

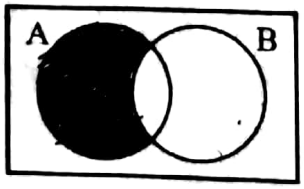


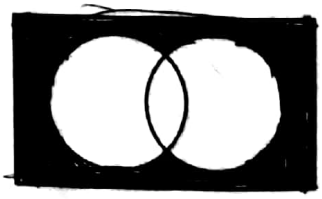

(A'UB)'

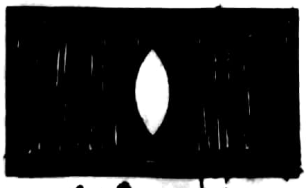



AnB

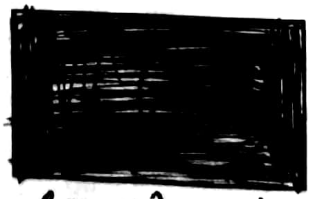




(A'nB)'



(AUB)U(A'UB')



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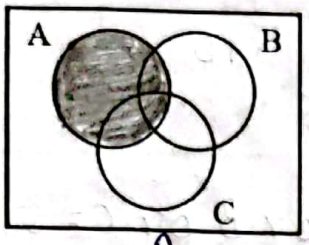
2) Use separate Venn Diagrams and shade the regions for the following:

(a) A (b) B' (c) $A \cup C$ (d) $B' \cup C$

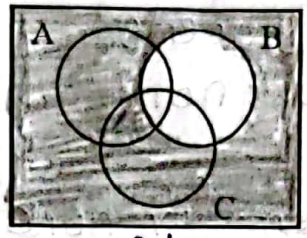
(e) $A \cup B' \cup C$ (f) $B' \cup C' \cup A$ (g) $(A \cup B) \cap C'$

(h) $(A' \cup B)' \cup C$ (i) $B \cap A \cap C$ (j) $(C' \cap B) \cup A$

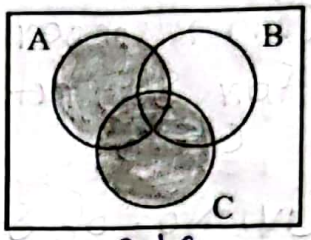
**Solution :**



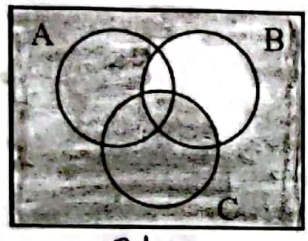

A



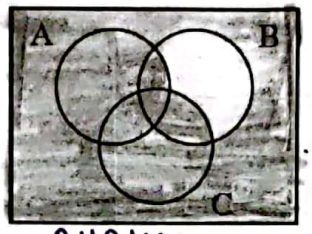

B'



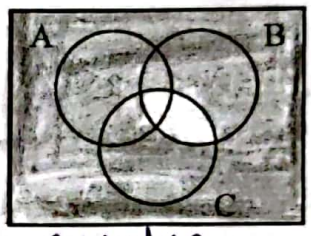

$A \cup C$



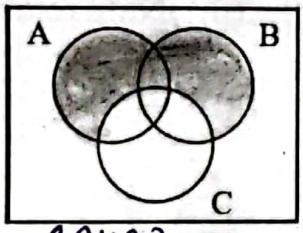

$B' \cup C$



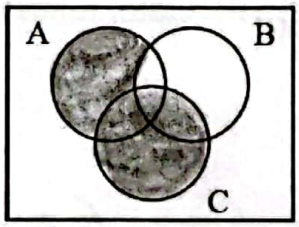

$A \cup B' \cup C$



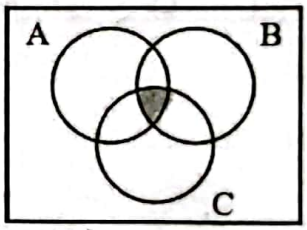

$B' \cup C' \cup A$



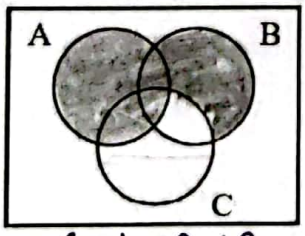

$(A \cup B) \cap C'$




$(A' \cup B)' \cup C$




$B \cap A \cap C$




$(C' \cap B) \cup A$

3) In a group of 50 students, 20 study subject A, 25 study subject B and 20 study subject C. 10 study both A and B, 5 study both B and C, 7 study both A and C. 2 study all three subjects.

(a) show this information on Venn diagram

(b) find the number of students who study

(i) A only (ii) B or C

(iii) A but not C (iv) none of A , B or C.



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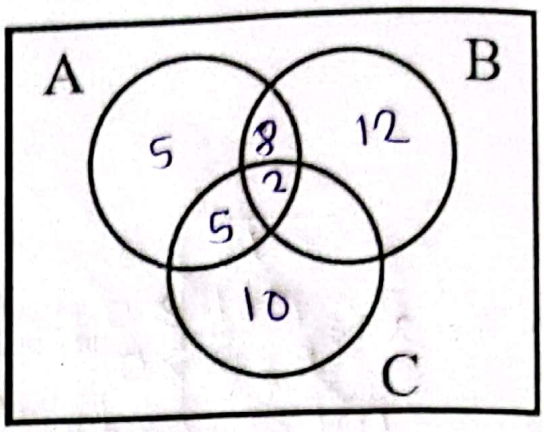
**Solution :**

(b)(i) Number of students for Subject A only = 5

(ii) Number of Students for subject B or C = n(B∪C) = 40

(iii) Number of students for Subject A but not C = 13



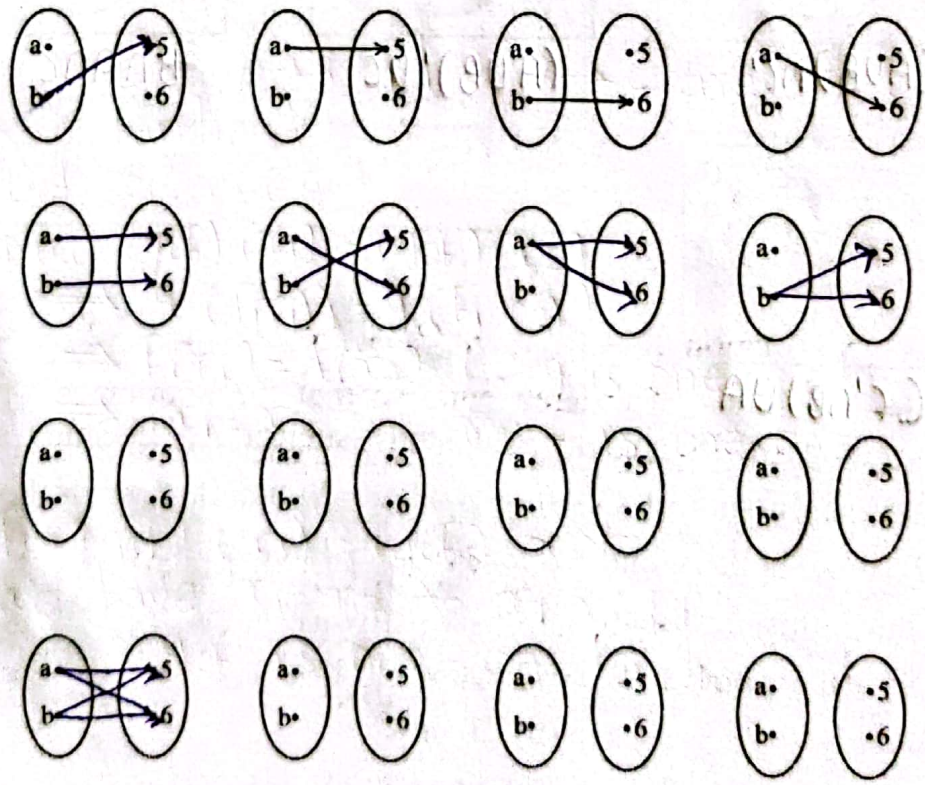

(iv) No. of student for none A or B or C = 5

4) Show all possible relations from A = {a, b} to B = {5, 6} using Arrow diagram. Find among them the relations that are :

(i) One-One relation (ii) Onto relations (iii) Null relation

(iv) Universal relation.

**Solution :**



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